BharatGov Access — National Observatory

GIGW 3.0 Compliance & Web Infrastructure Quality Audit Report (v2.5.0)

2,390	92.3 / 100	31.4%	28.8%	100%
Monitored Portals (Central/State/Districts)	Avg WCAG AA Score (Automated axe-core)	GIGW Toolbar Adoption (Font & Contrast Toggle)	Multilingual Reach (Scheduled Regional Scripts)	HTTPS Enforcement (21.7% HSTS Strict)

1. Executive Overview & Empirical Methodology

This observational audit synthesizes deterministic measurements from **2,390 public-facing Indian government websites** across 4 administrative tiers: Central Ministries (437), State/UT Portals (1,057), District Administrations (867), and Urban Local Bodies (29). Each domain was evaluated through multi-page traversal (Homepage, About, Contact, Citizen Services, Circulars) using headless Playwright Chromium, axe-core 4.8, Lighthouse Core Web Vitals, and SSL/TLS cipher suites under strict safety governance (≤ 1 req/sec/domain).

2. Tier-Wise Web Infrastructure Benchmark

Governance Tier	Sample (N)	Avg WCAG	Toolbar Rate	Multilingual	Median LCP	HSTS Strict
Central Ministries	437	94.2	42.1%	68.4%	3.2s	38.6%
State / UT Portals	1,057	91.8	31.4%	38.2%	3.7s	24.1%
District Administration	867	93.1	26.5%	19.5%	4.1s	14.8%
Local Bodies (ULBs)	29	88.5	17.2%	13.8%	4.4s	10.3%

3. Key Policy Findings & Compliance Disparities

- **GIGW 3.0 Toolbar Gap:** Only 31.4% of portals provide high-contrast toggles or font resizing (A+/A-), disproportionately impacting visually impaired citizens.
- **Screen Reader Accessibility:** Missing image alt text remains the single most prevalent defect, occurring on 68.2% of state department portals.
- **District Language Disparity:** While Central portals exhibit 68.4% bilingual coverage (Hindi/English), district portals achieve only 19.5% support for local scheduled languages (Tamil, Telugu, Bengali, Kannada, Marathi).
- **Security Hygiene Deficit:** Although HTTPS is enforced on 100% of portals, only 21.7% implement HSTS headers and 19.2% declare Content-Security-Policy (CSP), leaving domains vulnerable to downgrade and cross-site scripting attacks.

4. State Web Infrastructure Maturity Matrix

Composite benchmark derived from WCAG AA compliance, Core Web Vitals speed index, GIGW utility adoption, and mobile responsive layout across audited state properties:

Rank	State / UT	Portals	Maturity Score	GIGW Toolbar	Regional Language	Leading Architecture
1	Kerala	34	88.4 / 100	64.7%	82.4%	Drupal / Custom SPA
2	Tamil Nadu	42	85.2 / 100	57.1%	78.6%	S3WaaS / WordPress
3	Karnataka	38	83.7 / 100	52.6%	73.7%	NIC CMS / Drupal
4	Maharashtra	48	81.9 / 100	47.9%	66.7%	S3WaaS / WordPress
5	Gujarat	36	79.5 / 100	44.4%	58.3%	S3WaaS Framework
6	Uttar Pradesh	76	76.2 / 100	39.5%	46.1%	S3WaaS / Dynamic IIS
7	Assam	67	74.8 / 100	34.3%	37.3%	S3WaaS Framework
8	Rajasthan	49	72.1 / 100	30.6%	32.7%	Drupal / WordPress
9	Madhya Pradesh	53	69.5 / 100	26.4%	28.3%	Legacy Dynamic Portal
10	Bihar	45	66.8 / 100	22.2%	24.4%	Static HTML / ASP.NET

5. National Standardization & Policy Recommendations

1. **Mandatory S3WaaS Migration:** Empirical analysis demonstrates that districts utilizing NIC's Secure, Scalable and Sugamya Website as a Service (S3WaaS) platform achieve 34% higher accessibility scores and 52% lower WCAG AA critical violation counts than legacy custom CMS instances.
2. **Regional Language Mandate for District Portals:** District administrations must be supplied standardized bilingual CMS templates featuring automated localization for official scheduled languages in their corresponding state.
3. **Security Header Enforcement:** MeitY and CERT-In should issue binding guidelines requiring immediate inclusion of Strict-Transport-Security (max-age=31536000) and strict Content-Security-Policy across all NIC/State data center load balancers.
4. **Automated CI/CD Accessibility Auditing:** Deploy continuous axe-core linting in government web deployment pipelines to eliminate broken image alt attributes and unlabeled form inputs prior to production deployment.

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